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# Overriding an IMS Database Name in a Workflow

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# Abstract

You can create a PowerCenter workflow that remotely overrides the IMS database name that is input to a bulk data movement session. This article describes how to create the JCL, UNIX scripts and data files, PowerExchange data maps, and PowerCenter mappings, tasks, and workflow to accomplish this task.

# Supported Versions

- PowerExchange 10.0 or later

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# Scenario

You are a PowerExchange and PowerCenter developer who has created a bulk data movement session to move IMS data to a target system. You want to override the names of the IMS database files that are input to the bulk data movement session, so that you can use the session with a different database. However, you do not have access to the z/OS source system.

To accomplish this goal, you can create a PowerCenter workflow that includes the following tasks:

- A session that copies a data file from the Integration Service UNIX machine to a temporary data set on z/OS. The file contains JCL DD statements that define the names of the IMS database files.
- A command task that runs the DTLREXE utility on the Integration Service UNIX machine to remotely execute the IEBGENER utility on z/OS. IEBGENER copies the temporary data set to the IMSDATA data set. DTLREXE then executes IEBGENER again to copy the contents of IMSDATA to a result data set and copies the result data set to the Integration Service machine, so that you can verify the results.
- The bulk data movement session that you already created. This session uses a netport job to extract data from IMS. The JCL for this job contains an INCLUDE statement for the IMSDATA data set.

Before you create the workflow elements, you must configure PowerExchange, create a data map, create files on UNIX, and tailor the JCL on z/OS. This article describes all these steps.

This article assumes that you have already set up a bulk data movement session to move IMS data to a target. To create this session, you have performed the following tasks:

- Tailored the JCL for a netport job.

- Edited the DBMOVER configuration files on the source, target, PowerExchange Navigator and PowerCenter client, and Integration Service machines.
- Created data maps for the source and, if necessary, target systems.
- Created the PowerCenter mapping, defined connections, and created the bulk data movement session.

## Task Flow

To override an IMS database name in a workflow, complete the following tasks:

1. Edit the DBMOVER files.
2. Create a data map.
3. Create the data file that contains JCL DD statements on the Integration Service machine.
4. Tailor the DTLREXE JCL.
5. Create a UNIX script that calls DTLREXE.
6. Create a mapping and session that copies the data file from UNIX to a temporary data set on z/OS.
7. Create a command task that calls DTLREXE to copy the temporary data set to the IMSDATA data set and then to a result set.
8. Create a workflow that includes the tasks that you created in Steps 6 and 7 and the bulk data movement session that you created previously.
9. Run the workflow.

## Step 1. Configure DBMOVER Configuration Files

Include LISTENER or NODE statements in the DBMOVER configuration files. You may have already included some of these statements when you set up the bulk data movement session.

- On the PowerExchange Navigator and PowerCenter client machine, include NODE statements that identify the PowerExchange Listener ports on the following machines:
  - PowerCenter Integration Service machine
  - z/OS source machine
  - Target machine of the bulk data movement mapping

  To identify separate ports for standard PowerExchange Listener requests and netport job requests on the z/OS source machine, define separate NODE statements.
- On the PowerCenter Integration Service machine, include NODE statements that identify the PowerExchange Listener ports on the following machines:
  - z/OS source machine
  - Target machine of the bulk data movement mapping

  To identify separate ports for standard PowerExchange Listener requests and netport job requests on the z/OS source machine, define separate NODE statements.
- On the z/OS source machine, include the following statements:
  - LISTENER statement for standard PowerExchange Listener requests
  - LISTENER statement for netport job requests
  - NETPORT statement for netport job requests
- On the target machine of the bulk data movement session, include a LISTENER statement.

## Step 2. Create a Data Map to Copy the Data File

In the PowerExchange Navigator, create a data map for the data file that you copy from UNIX to z/OS. You can use the same map for the UNIX and z/OS files and specify a file name override in the session properties for the UNIX file. Alternatively, you can create two separate data maps.

You will import the data map into PowerCenter to create source and target definitions. You will then use these definitions in a mapping that copies the data file from UNIX to z/OS.

1. In the PowerExchange Navigator, click **Add** > **Data Map**.
2. In the **Name** dialog box, enter the following information:
   - In the **Schema Name** box, enter a user-defined schema name, such as SEQ.
   - In the **Access Method** list, select **SEQ**.
   - In the **Data Map Name** box, enter a user-defined data map name, such as DTLREXE.
   - Clear **Import Record Definitions**.
3. Click **Next.**

   The **SEQ Access Method** page appears.
4. In the **File** field, enter the name of the temporary data set to which the workflow will copy contains the IMS database file names.

   For example, enter *hlq*.**STD.DTLREXE**.

   This data set name must match the DSN value in the SYSIN DD of the JCL on z/OS.
5. In the **Record Format** box, select **Fixed**, with a **Size** of **80.**
6. Click **Finish**.
7. In the **Resource Explorer** pane, right-click the data map name, and select **Add Record**.
8. In the **Add Record** dialog box, enter a record name, such as `ctrlcard`. Then click **OK**.
9. In the **Record** dialog box, right-click the record, and select **Add Field at End**.

   You must add a field to hold the data file names.

   The **Add Field** dialog box appears.
10. On the **Name** tab, enter the following information:
    - Enter a field name, such as `ctrlcard`.
    - In the **Field Type** list, select CHAR.
    - In the **Length** field, enter 80.
11. Click **OK**.
12. In the **Resource Explorer** pane, right-click the data map name and select **Add Table**.
13. In the **Add Table** dialog box, enter a table name, such as `ctrlcard`. Then click **OK**.
14. In the **Resource Explorer** pane, right-click the table and select **Properties**.

    The **Table Properties** dialog box appears.
15. Click **File** > **Save**.
16. Click **File** > **Send to Remote Node**.
17. In the **Data Map Remote Node** dialog box, enter the connection information for the remote z/OS system. Then click **OK**.

    In the **Location** list, select the node name for the z/OS system. You must have defined this node name in a NODE statement in the DBMOVER configuration file.

18. In the **Data Map Remote Node** dialog box, enter the connection information for the Integration Service machine. Then click **OK**.

    In the **Location** list, select the node name for the Integration Service machine. You must have defined this node name in a NODE statement in the DBMOVER configuration file.

For more information about creating data maps, see the *PowerExchange Navigator User Guide*.

## Step 3. Create the Data File on the Integration Service Machine

On the Integration Service machine, where the DTLREXE script will run, use a text editor to create a file that contains the IMS database file names.

1. In the file, enter the DD statements for the IMS database file names, such as the following statements:

```
//IMSSAMDB   DD DSN=hlq.IMST.IMSSAMDB,DISP=SHR
//IMSSAMIX   DD DSN=hlq.IMST.IMSSAMIX,DISP=SHR
```

2. Save the file to a file name and path of your choice, such as /home/*mydir*/imsovr.

## Step 4. Customize the DTLREXE JCL

On the z/OS system, customize the DTLREXE JCL. Include two EXEC statements to execute IEBGENER: first to copy the temporary data set to IMSDATA, then to copy IMSDATA to a result set. Use the DTLREXE member in the RUNLIB library as a starting point.

Coordinate with your z/OS system administrator to perform this task.

1. Copy the DTLREXE member in the RUNLIB under another name.

   This scenario uses the member name of *hlq*.V901.RUNLIB(IMSDBOVR).

2. In the IMSDBOVR member, insert JCL to perform the following tasks:

   - Execute the IEBGENER utility to copy the temporary data set to IMSDATA
   - Execute IEFBR14 to clear a result set from a previous run
   - Execute IEBGENER again to copy IMSDATA to a a result set.

   Insert these steps where the following comment section occurs:

```
//*-------------------------------------------------------
//*
//* <<<<< INSERT DESIRED STEPS HERE >>>>>
//*
//*-------------------------------------------------------
```

3. In the JCL, customize the DDs to point the correct data sets on your z/OS system.

For this scenario, the entire job appears as follows:

```
//userid JOB  'PWX INSTALL',MSGLEVEL=(1,1),MSGCLASS=X,
//       CLASS=A,REGION=0M
//LIBSRCH   JCLLIB ORDER=hlq.V901.RUNLIB
//*
//*
//*
//* MEMBER DTLREXE
//*
//INCS1 INCLUDE MEMBER=GENBULK
//*--------------------------------------------------------------------
//*  OVERRIDES IMS DATABASE NAME IN IMSDATA INCLUDE MEMBER OF NETPORT
//*
//*  JCL TEMPLATE FOR USE WITH DTLREXE PROG=SUBMIT.
//*
//*  THIS TEMPLATE CAN BE TAILORED TO SUIT A SITE'S SPECIFIC NEEDS
//*  WITH THE FOLLOWING RESTRICTION:
//*
```

```
//*  - THE START AND END STEPS MUST BE INCLUDED IF MODE=(JOB,WAIT)
//*    OR MODE=(JOB,TIMED) ARE SPECIFIED ON THE DTLREXE COMMAND.
//*
//*  INSERT THE STEPS DESIRED BETWEEN STEPS AND ENDERR
//*
//*-------------------------------------------------------------------
//*
//START    EXEC PGM=DTLNTS,PARM='"%STRTJOB"'
//STEPLIB  DD DSN=&HLQ..LOADLIB,DISP=SHR
//*-------------------------------------------------------------------
//*
//*
//* ------------------------------------------------------------------
//* OVERRIDES IMS DATABASE NAME IN IMSDATA INCLUDE MEMBER OF NETPORT
//* ------------------------------------------------------------------
//*
//ALLOCDBD EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSOUT   DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//SYSUT2   DD DSN=hlq.V901.RUNLIB(IMSDATA),DISP=SHR
//SYSUT1   DD DSN=hlq.STD.DTLREXE,DISP=SHR
//SYSIN    DD DUMMY
//*
//*
//* ------------------------------------------------------------------
//*   DELETE JWAITE.IMSREX
//* ------------------------------------------------------------------
//*
//BR14   EXEC PGM=IEFBR14
//DD1       DD DSN=hlq.IMSREX,
//             DISP=(MOD,DELETE,DELETE),
//             UNIT=3390,SPACE=(TRK,(0,0)),
//             DCB=(LRECL=80,RECFM=FB,BLKSIZE=3120)
//*
//ALLOCDBD EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSOUT   DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//SYSUT2   DD  DSN=hlq.IMSREX,DISP=(NEW,CATLG,DELETE),
//             SPACE=(TRK,(1,1)),UNIT=3390,
//             DCB=(LRECL=80,RECFM=FB,BLKSIZE=3120)
//SYSUT1   DD DSN=hlq.V901.RUNLIB(IMSDATA),DISP=SHR
//SYSIN    DD DUMMY
//*
//*
//*
//*-------------------------------------------------------------------
//         IF ((RC > 4) | (ABEND=TRUE)) THEN
//*
//ENDERR   EXEC PGM=DTLNTS,
//   PARM='"%ENDJOB" C 16'
//STEPLIB  DD DSN=&HLQ..LOADLIB,DISP=SHR
//SYSPRINT DD SYSOUT=*
//*
//         ELSE
//*
//ENDOK    EXEC PGM=DTLNTS,
//  PARM='"%ENDJOB"'
//STEPLIB  DD DSN=&HLQ..LOADLIB,DISP=SHR
//SYSPRINT DD SYSOUT=*
//         ENDIF
```

## Step 5. Create a Script that Calls DTLREXE to Submit the Remote JCL on z/OS

On the UNIX system, create a script or batch file that calls the DTLREXE utility to remotely submit the JCL with the EDMLUCTR execution step on z/OS.

This scenario uses a file name of /home/*mydir*/imsovr. Iinclude the following parameters and syntax:

```
dtlrexe prog=submit loc=zos_node_name fn=\"hlq.RUNLIB(IMSDBOVR)\"
mode=(job,timed) time=20 output=\"hlq.IMSREX\"
result=/home/mydir/IMSRSLT
```

The following table describes the DTLREXE parameters:

| Parameter | Description |
|---|---|
| prog | The DTLREXE program. This value must be submit to submit the remote JCL. |
| loc | Your z/OS node name, as defined in the NODE statement of the DBMOVER configuration file. |
| fn | The fully qualified name of the member on z/OS that contains the JCL. You must enclose this name with the backslash and escape character, that is \". |
| mode | A submit mode that is composed of the following values:<br>- **job**. Submits a job.<br>- **timed**. After submitting a job, DTLREXE waits for the job to complete. The wait period is specified in the time parameter. |
| time | Number of seconds to wait for the job to return results. This time interval begins when the job is submitted. |
| output | The fully qualified name of the member on z/OS that contains the IEBGENER output. You must enclose this name with the backslash and escape character, that is \". |
| result | The full path and file name of the result file on UNIX to which IEBGENER output is sent. |

For more information, see the *PowerExchange Utilities Guide*.

## Step 6. Create a Mapping and Session that Copies the Data File from UNIX to z/OS

In the PowerCenter Designer, create a mapping that maps the files that contain the DD statements that define the database file names. In the Workflow Manager, create a connection and session that copies the file on UNIX to a temporary data set on z/OS. A subsequent task in the workflow will copy the temporary data set to the IMSDATA data set.

When setting up the mapping, use the following guidelines:

- Import the data map for the DD statement file on UNIX to create a source definition.
- Import the data map for the DD statement temporary data set on z/OS to create a target definition.
- Use these source and target definitions in the mapping.

When setting up the session, use the following guidelines:

- Use a PWX NRDB Batch application connection.

- You can create one data map for the target control statement data set on z/OS and use it to also create the source definition. In Task Developer, open session properties. On the **Mapping** tab, click **Sources view**. In the **Properties** settings, set the **File Name Override** attribute to point to the data file on UNIX.

For more information, see *PowerExchange Interfaces for PowerCenter*.

## Step 7. Create a Command Task that Runs the DTLREXE Utility

In the Task Designer, create a task that runs the DTLREXE job that you defined in Step 4. This job executes IEBGENER to copy the temporary data set containing the DD statements to the IMSDATA data set. The job then executes IEBGENER again to copy the contents of IMSDATA to a result data set on z/OS and a result file on UNIX, so that you can verify the contents.

1. In the Workflow Manager, open the Task Designer, and select **Tasks > Create**.
2. In the Create Task dialog box, select **Command** as the task type.
3. Enter a name for the task, and click **Create** and then **Done**.
4. Double-click the task, and select the **Commands** tab in the **Edit Task** dialog box.
5. At **Command**, enter the path and file name of the script that you created in Step 5, such as:

   ```
   /home/mydir/imsovr
   ```
6. Select **Repository > Save**.

## Step 8. Create a Workflow

In the Workflow Designer, create a workflow that includes the tasks that you have created.

1. Open the Workflow Designer and select **Create > Workflow**.
2. In the Create Workflow dialog box, enter a name, and click **OK**.
3. Drag the following tasks into the workplace:
   - The session that you created in Step 6
   - The task that you created in Step 7
   - The bulk data movement session that you created previously

   Select **Tasks > Link Task** to link each task to the next one.
4. Select **Repository > Save**.

## Step 9. Run the Workflow

To start the workflow from the Workflow Manager, select **Workflow > Start Workflow**.

The first time that you run the workflow, verify that the /home/*mydir*/IMSRSLT result file on the Integration Service machine was copied correctly. If you find a problem with the file, you can stop the workflow manually.

If you run multiple workflows that access the IMSDATA data set, you must ensure that they do not run concurrentlly, so that the contents of IMSDATA are valid for each workflow.

## Author

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